Thermodynamic Results Summary

Turbofan Engine V1

Preliminary Thermodynamic Performance Summary

1. Engine Configuration

Parameter	Value
Engine Type	Two-Spool High-Bypass Turbofan
Bypass Ratio	5.0
Core Mass Flow	20.329 kg/s
Bypass Mass Flow	100.000 kg/s
Total Air Mass Flow	120.000 kg/s
Operating Condition	Static Sea-Level

2. Main Thermodynamic Results

Parameter	Value
Combustor Exit Temperature	1400 K
High-Pressure Turbine Exit Temperature	1003.25 K
Low-Pressure Turbine Exit Temperature	637.39 K
Core Nozzle Exit Velocity	390.14 m/s
Bypass Nozzle Exit Velocity	284.74 m/s

3. Engine Performance Summary

Performance Parameter	**Value**
Core Thrust	7.93 kN
Bypass Thrust	28.47 kN
Total Net Thrust	36.41 kN
Core Thrust Contribution	21.79 %
Bypass Thrust Contribution	78.21 %
Specific Thrust	303.38 N/(kg/s)
Fuel Mass Flow Rate	0.329 kg/s
TSFC	32.53 kg/(kN·h)

4. Key Engineering Observations

- Two-spool high-bypass turbofan thermodynamic cycle successfully completed.
- Core and bypass streams were analysed independently before thrust integration.
- Both exhaust nozzles operated under **unchoked** conditions at the assumed static sea-level operating point.
- Approximately **78%** of the engine thrust was generated by the bypass stream, consistent with modern high-bypass turbofan architectures.
- Total estimated engine thrust was **36.41 kN** under the assumed design conditions.
- GNU Octave was used to implement and verify all thermodynamic calculations.

5. Analysis Assumptions

- One-dimensional steady-flow analysis
- Constant specific heats
- Ideal gas behaviour
- Complete combustion
- Fixed component efficiencies
- Static sea-level operation
- Negligible pressure thrust for perfectly expanded nozzles

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